How to Train Your C. Elegans: Inducing Thermotactic Behavior in a Digitally Simulated Worm

Adam Cheong

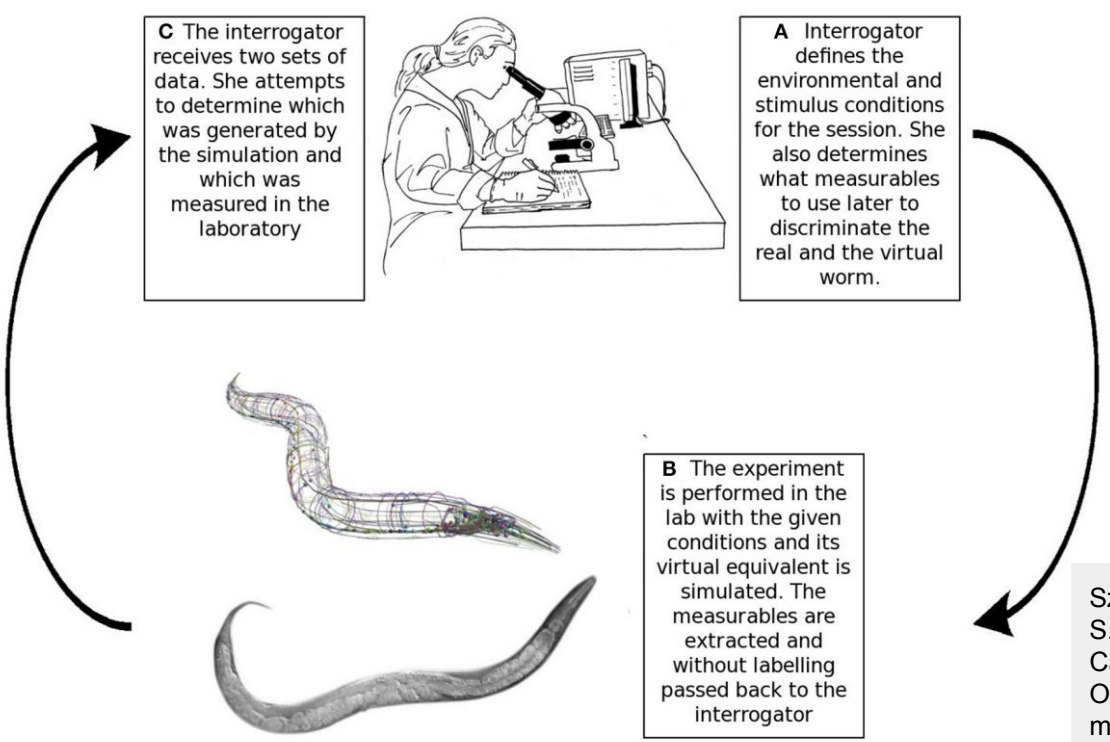
Dear Dr. Yetman,

Can't make Nov 5th for Regeneron but will try to finish before GGSF (Feb 28, 2027).

Thanks,
Adam



Background: A “Turing Test” for In Silico Biology

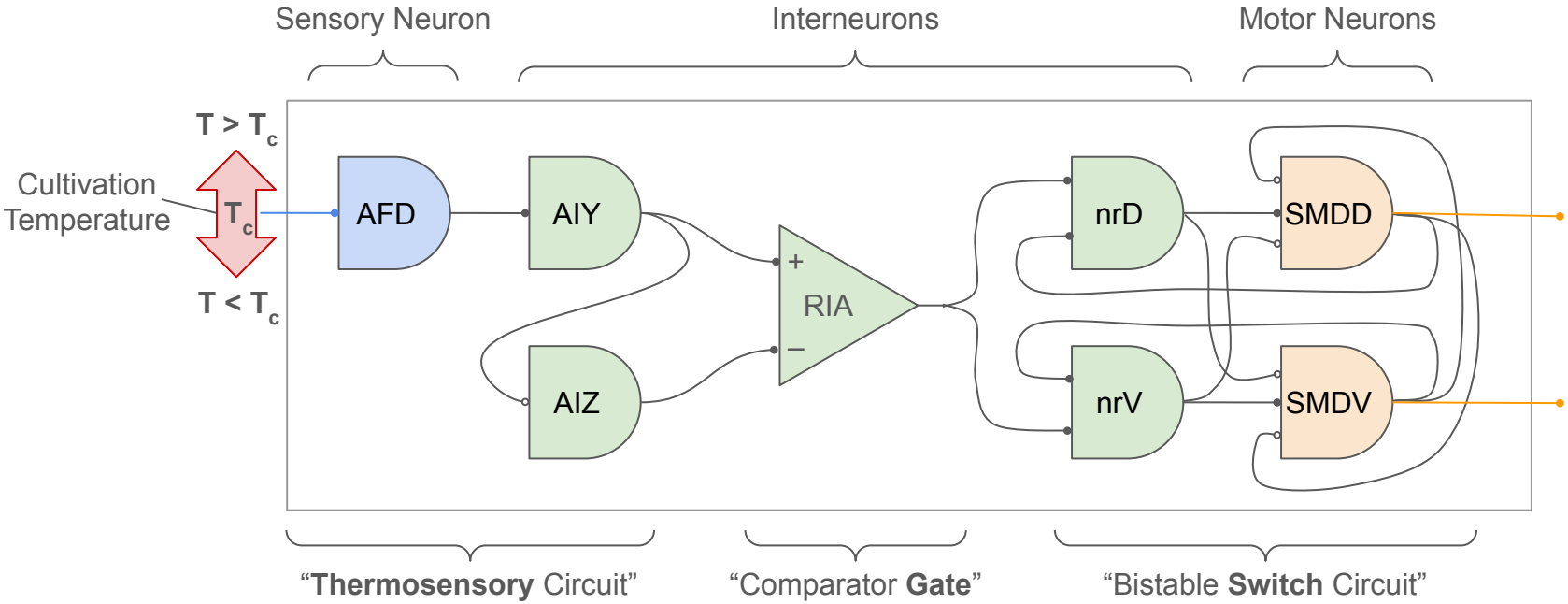


Szigeti, B., Gleeson, P., Vella, M., Khayrulin, S., Palyanov, A., Hokanson, J., Currie, M., Cantarelli, M., Idili, G., & Larson, S. (2014). OpenWorm: an open-science approach to modeling Caenorhabditis elegans. Frontiers in Computational Neuroscience, 8.

“Neurons are Just Wires” — A Falsifiable Hypothesis

- A conundrum: “How is it that the OpenWorm project, running since 2011 with a fully mapped connectome and a detailed biomechanical model of the worm's body, has not produced a digital *C. elegans* that crawls towards food the way a real animal does?”
- This begs the question: “Does the connectomic gap come from 15 years of inadequate software modeling or 40 years of mapping the wrong layer of the brain?”
- We know the connectome is *necessary* information for understanding a nervous system — but is it *sufficient?* What could we have missed?
- E.g., basal cognition (hallmarks of intelligence outside of neurons): retained memory of regenerated planaria, sea slug RNA as memory storage, bioelectricity in cells, etc?
- Our research goal: A clean simulation of *C. elegans* with the connectome, reproducing biomechanical behavior such as thermotaxis (as a first baby step), so as to falsify the "neurons are just wires, no behavior from wiring diagram” hypothesis.
- Therefore: “neurons are wires that encode behavioral response to environmental stimuli.”

Deciphering Behavioral Circuit at the Connectome Level

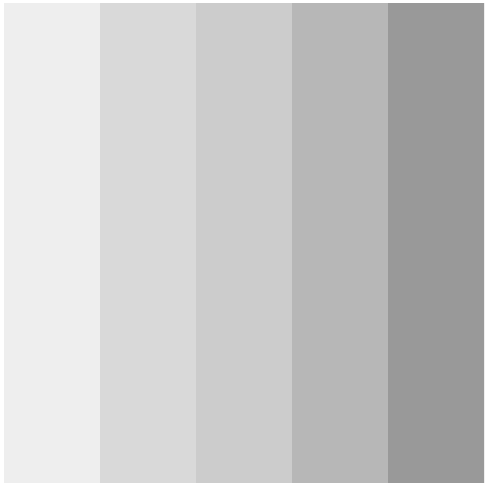


Ouellette, M.-H., Desrochers, M. J., Gheta, I., Ramos, R., & Hendricks, M. (2018). A gate-and-switch model for head orientation behaviors in caenorhabditis elegans. Eneuro, 5(6), ENEURO.0121–18.2018.

Methods: (1) Introduce Temperature Gradient into Aquarium

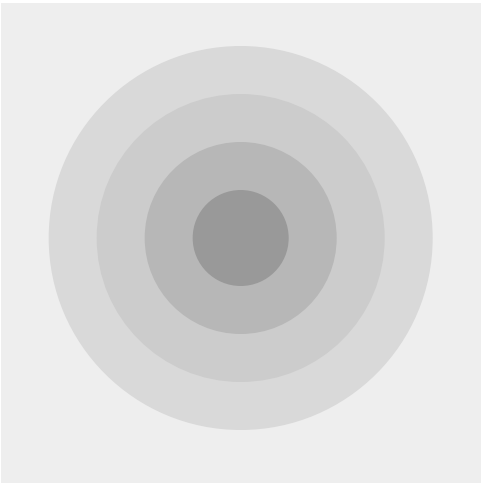
(a) Uniform (T_c)

Temperature Control (Default)



(b) Symmetric

“Vertical Bands”



(c) Asymmetric

“Concentric Rings”

Methods: (2) Reproduce AFD-AIY Transfer Characteristics

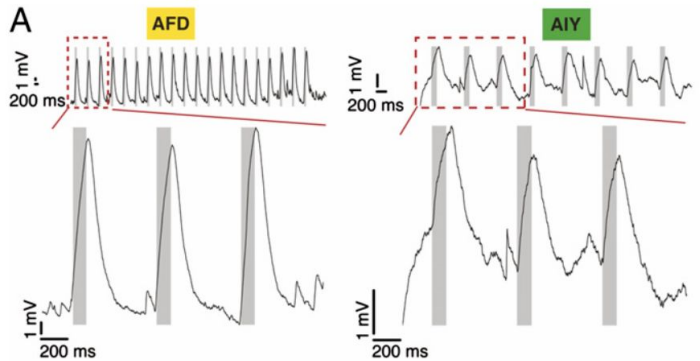
Table 1. Orientation of thermotaxis is determined by connectivity of AFD and AIY.

Genotype	AFD-AIY connectivity	Probability of AIY response	Behavior
ttx-3, AIY ablation, *dgk-1*, AFD::caPKC-1	AFD AIY		Negative thermotaxis
Wild type ($T>T_c$)	AFD → AIY		
Wild type ($T<T_c$)	AFD → AIY		Positive thermotaxis
pkc-1/ttx-4	AFD → AIY		

More connectivity induces higher probability of responses of AIY. Lower probability of AIY responses directs worms toward cooler side, and vice versa. Genotype: overexpression of constitutively-active PKC-1 into AFD (AFD::caPKC-1).

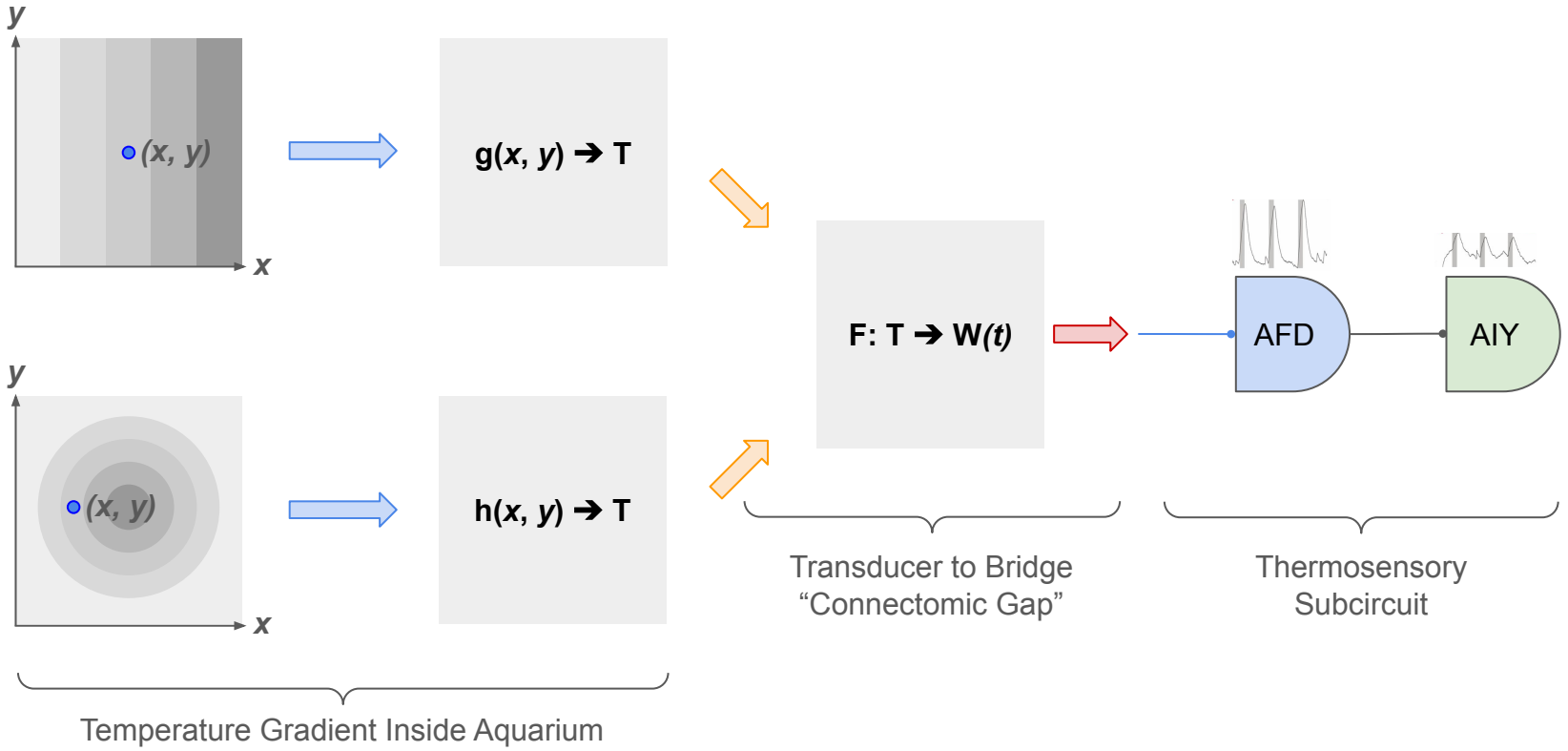
Takeishi, A., Takagaki, N., & Kuhara, A. (2020). Temperature signaling underlying thermotaxis and cold tolerance in caenorhabditis elegans. Journal of Neurogenetics, 34(3–4), 351–362.

"The fact that the AIY response is tonic, shows no adaptation, and is frequency independent over the range of inputs explored would suggest that the role of this synapse is to provide AIY with a faithful scaled-down version of temperature changes tracked by AFD."

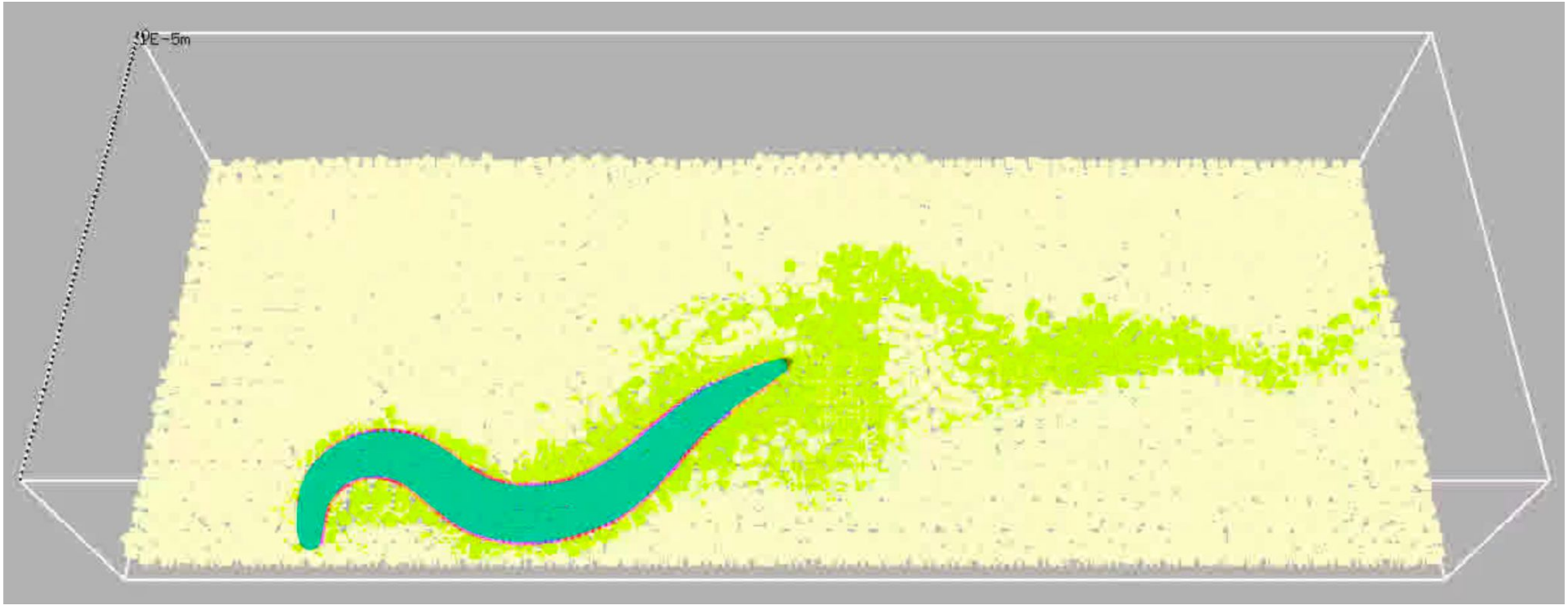


Narayan, A., Laurent, G., & Sternberg, P. W. (2011). Transfer characteristics of a thermosensory synapse in caenorhabditis elegans. Proceedings of the National Academy of Sciences of the United States of America, 108(23), 9667–9672.

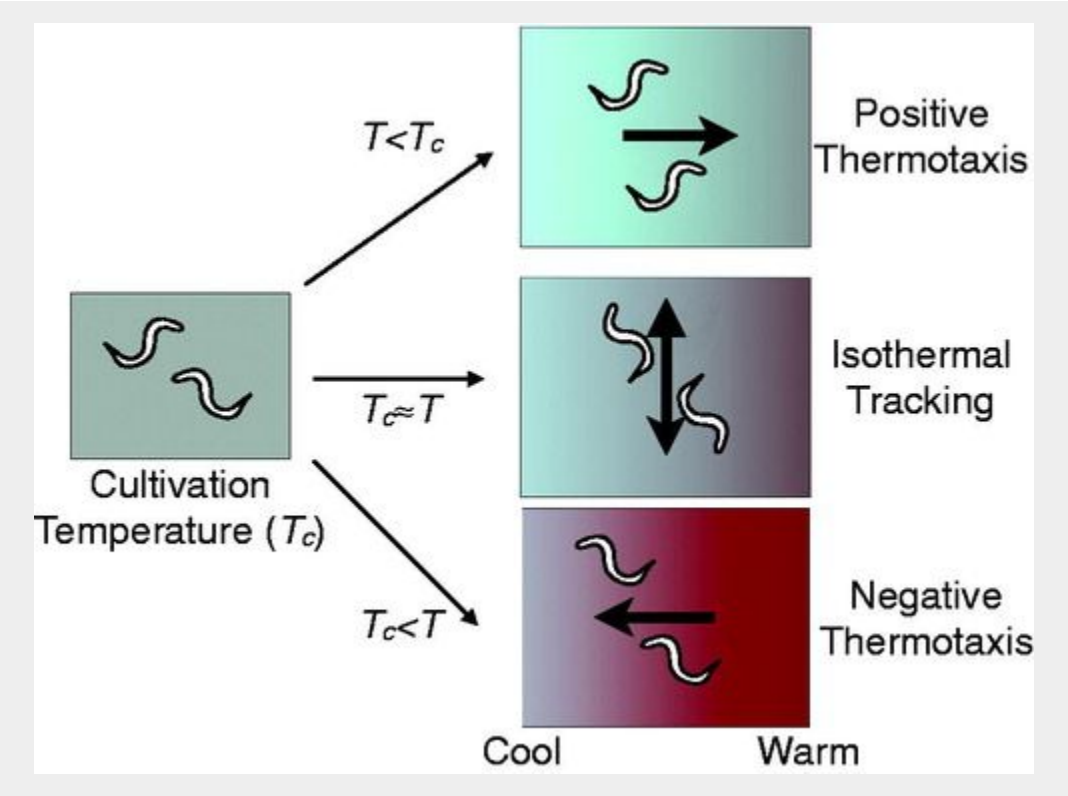
Methods: (3) Construct Transducer for AFD Neurons



Methods: (4) Release Sensitized Digital Worm into Aquarium



Results: Visual Confirmation of Thermotactic Behaviors



Conclusion: “Connectome Encodes Behavioral Response”

- There is no environmental stimuli in OpenWorm. Its simulated aquarium acts like a virtual “sensory deprivation tank” for the biomechanical worm model.
- I’ve found a way to introduce temperature gradient into the aquarium so as to study thermotaxis in a sensitized digital worm equipped with transducers.
- I hope to show that “neurons are wires that encode behavioral response to environmental stimuli” by demonstrating thermotaxis using OpenWorm.
- Thus proving the connectome already encodes behavioral response, and no additional training (surprise!) is required for thermotaxis in a digital worm.

Our Approach: Take *Educated Baby Steps* with Digital Worm

A recent paper by 37 co-authors outlined a new plan:

“Use genetic imaging technology to activate each neuron in the worm’s nervous system one by one, measuring its effect on the other 301. Repeated hundreds of thousands of times in parallel experiments, this methodical process should hoover up enough data to give the computational folks, finally, something to work with—enough, even to ‘reverse engineer’ the worm completely.”

It is estimated that the proposed plan will:

— take up to 10 years,
— cost tens of millions of dollars, and
— require 100,000m to 200,000 real-life worms.

Evans, C. L. (2025, March 26). The worm that no computer scientist can crack. Wired.

My proposed research plan takes an incremental approach studying one chosen behavior at a time:

1. Decipher behavioral circuit at the connectome level informed by background review of literature;
2. Introduce environmental stimuli into the simulated aquarium as a well-designed experimental setup;
3. Figure out how to reproduce transfer characteristics of signals representing environmental stimuli so behavioral circuit operates as intended by nature;
4. Construct transducers as sensory front-ends to neural pathways responsible for target behavior;
5. Release sensitized digital worm into the simulated aquarium—now set up for simulation experiments.
6. Observe behaviors emerge in silico from the digital worm, matching its real-world animal counterpart.

Randi, F., Sharma, A. K., Dvali, S., & Leifer, A. M. (2023). Neural signal propagation atlas of caenorhabditis elegans. Nature, 623(7986), 406–414.

Project Summary

How is it that the OpenWorm project, running since 2011 with a fully mapped connectome and a detailed biomechanical model of the worm's body, has not produced a digital *C. elegans* that crawls towards food the way a real animal does? Does this connectomic gap come from 15 years of inadequate software modeling? Or 40 years of mapping the wrong layer of the brain? What could we have missed?

Our research goal is to show that behavioral circuits at the connectome level are sufficient to reproduce real-world animal behavior. In fact, the AFD-AIY connectivity along the worm’s thermosensory pathway is known to drive thermotactic behavior in response to a temperature gradient.

To bridge the “connectomic gap” between the biomechanical worm model and its simulated aquarium, we construct thermosensory front-ends (called “transducers”) that faithfully reproduce transfer characteristics of signals across a temperature gradient so the worm’s thermosensory circuit operates as intended.

We run a series of biomechanical simulations of this newly sensitized digital worm inside the simulated aquarium, now set up with a well-calibrated temperature gradient. We seek visual confirmation of thermotactic behaviors (e.g., positive thermotaxis, negative thermotaxis, isothermal tracking) emerging in silico from the digital worm, matching its real-world animal counterpart.